R89/B_muon_hybrid | Locked full18 results

Same unique VAL checkpoint on VAL/TEST/LIB; native/FP64 ind; historical NSS weights disclosed.

epoch 25 | update 36000 | seed2 | fresh initialization

Data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
VAL ALL	71.2481	66.2037	11.416294	11.427978	11.371692	-0.1972	+0.3904	0.4007
TEST ALL	71.4198	66.1802	11.373470	11.385299	11.329932	-0.2014	+0.3817	0.3896

Dataset	n	NSS Top1 (%)	R89 Top1 (%)	NSS cost	R89 cost
TSP TEST	4500	45.888889	47.111111	7.970400719	7.970897718
CVRP TEST	4500	45.977778	48.644444	18.004802368	18.010343889
TSPLIB	49	20.408163	26.530612	8.217907074	8.202507382
CVRPLIB	100	70.000000	65.000000	67.101111717	67.237737150

NSS: R79 weights trained on historical70114 TRAIN, selected using historical VAL; not retrained on new90k.

NSS TEST predictions were freshly computed on these4500 instances; old3895 TEST metrics are not reused.

SBS is fitted on the new81000 TRAIN only. Oracle is the per-instance legal-pool minimum.

ALL is the equal-problem mean; mixed raw cost is descriptive. TEST informed prior research.

Method selection and true winner use separate colors on the same axes; Top1 is per-instance accuracy.

Selection head: original_score_mlp_maximin | objective: original native CE

R89/B_muon_hybrid | VAL: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.2481	66.2037	11.416294	11.427978	11.371692	-0.1972	+0.3904	0.4007	per-problem TRAIN SBS
TSP	46.9111	33.1111	8.025300	8.045354	7.990533	-0.2493	+0.4351	0.4231	T2T
CVRP	47.2222	18.7111	18.158686	18.269377	18.064087	-0.6059	+0.5237	0.6986	RELD_CVRP
ATSP	77.2444	47.4889	1.638565	1.676905	1.632471	-2.2863	+0.3733	0.3716	UNICO_MatPOENet
OVRP	41.2889	35.7556	7.461803	7.469579	7.415777	-0.1041	+0.6207	0.6321	MoSES_RF
VRPB	81.2000	80.8444	9.354460	9.354876	9.318750	-0.0044	+0.3832	0.3894	reld_mt
VRPL	38.1556	33.0889	12.431376	12.437959	12.383444	-0.0529	+0.3871	0.3940	MoSES_RF
VRPTW	80.1333	79.9556	18.818981	18.819431	18.766139	-0.0024	+0.2816	0.2975	reld_mt
OVRPTW	83.8667	83.7778	10.142515	10.143730	10.116859	-0.0120	+0.2536	0.2518	reld_mt
OVRPB	88.5333	88.5111	6.357409	6.357254	6.341078	+0.0024	+0.2575	0.2714	reld_mt
OVRPL	42.0000	35.9111	7.436879	7.445709	7.390416	-0.1186	+0.6287	0.6421	MoSES_RF
VRPBL	78.9556	78.5111	9.367927	9.371523	9.328366	-0.0384	+0.4241	0.4255	reld_mt
VRPBTW	78.6889	78.4222	19.944345	19.950430	19.851084	-0.0305	+0.4698	0.4506	reld_mt
VRPLTW	78.8000	78.7556	18.765349	18.765871	18.707843	-0.0028	+0.3074	0.3217	reld_mt
OVRPBL	89.4000	89.2000	6.368755	6.370093	6.352824	-0.0210	+0.2508	0.2629	reld_mt
OVRPBTW	83.5778	83.4444	10.661856	10.661098	10.623245	+0.0071	+0.3635	0.3418	reld_mt
OVRPLTW	83.6222	83.4889	10.156730	10.157423	10.129818	-0.0068	+0.2657	0.2693	reld_mt
VRPBLTW	79.6000	79.4000	19.792445	19.797044	19.706420	-0.0232	+0.4365	0.4261	reld_mt
OVRPBLTW	83.2667	83.2889	10.609915	10.609955	10.571302	-0.0004	+0.3653	0.3426	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R89/B_muon_hybrid | TEST: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.4198	66.1802	11.373470	11.385299	11.329932	-0.2014	+0.3817	0.3896	per-problem TRAIN SBS
TSP	47.1111	34.5333	7.970898	7.989621	7.937593	-0.2343	+0.4196	0.4115	T2T
CVRP	48.6444	17.9333	18.010344	18.124983	17.918509	-0.6325	+0.5125	0.6671	RELD_CVRP
ATSP	76.9111	47.2444	1.635238	1.674207	1.629441	-2.3276	+0.3557	0.3561	UNICO_MatPOENet
OVRP	40.4889	33.6444	7.446506	7.455750	7.400487	-0.1240	+0.6218	0.6329	MoSES_RF
VRPB	80.6889	80.4000	9.348779	9.351810	9.315298	-0.0324	+0.3594	0.3647	reld_mt
VRPL	38.3111	32.2667	12.348699	12.356138	12.300958	-0.0602	+0.3881	0.3945	MoSES_RF
VRPTW	81.1778	80.8444	18.688625	18.690545	18.637408	-0.0103	+0.2748	0.2857	reld_mt
OVRPTW	83.9333	83.9333	10.092670	10.093063	10.068135	-0.0039	+0.2437	0.2348	reld_mt
OVRPB	88.2222	88.1333	6.369608	6.370826	6.352225	-0.0191	+0.2736	0.2829	reld_mt
OVRPL	41.6222	34.6000	7.463239	7.472542	7.418033	-0.1245	+0.6094	0.6173	MoSES_RF
VRPBL	79.4667	79.4222	9.394811	9.396379	9.355967	-0.0167	+0.4152	0.4132	reld_mt
VRPBTW	78.8000	78.7556	19.667310	19.665725	19.577666	+0.0081	+0.4579	0.4560	reld_mt
VRPLTW	79.4222	79.2444	18.801197	18.803136	18.745143	-0.0103	+0.2990	0.3163	reld_mt
OVRPBL	89.2667	89.2889	6.375811	6.375741	6.359310	+0.0011	+0.2595	0.2705	reld_mt
OVRPBTW	84.9333	84.8222	10.658950	10.659591	10.624748	-0.0060	+0.3219	0.2948	reld_mt
OVRPLTW	83.0000	82.8667	10.085385	10.086112	10.059001	-0.0072	+0.2623	0.2617	reld_mt
VRPBLTW	79.8667	79.7333	19.702640	19.707068	19.613284	-0.0225	+0.4556	0.4369	reld_mt
OVRPBLTW	83.6889	83.5778	10.661758	10.662149	10.625574	-0.0037	+0.3405	0.3167	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R89/B_muon_hybrid | LIB and locked provenance

Current R89 factory, same buffers and selected checkpoint; no solver execution or LIB tuning.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
TSPLIB	26.5306	10.2041	8.202507	8.240435	8.142167	-0.4603	+0.7411	0.7525
CVRPLIB	65.0000	45.0000	67.237737	67.245364	66.945093	-0.0113	+0.4371	0.5389

SHA256: 41fa4ad8b590fa44e8949b114156d714d7ee2f2d31c847658a26654993057260

Unique selection: strict minimum full VAL macro_vs_sbs_pct with common new81000 TRAIN-fitted SBS; first wins ties

Inference: eval mode, fixed physical-node0 TSP slot3 role, original R77 score MLP/pair differences/maximin; stable ascending-global-solver-ID tie break

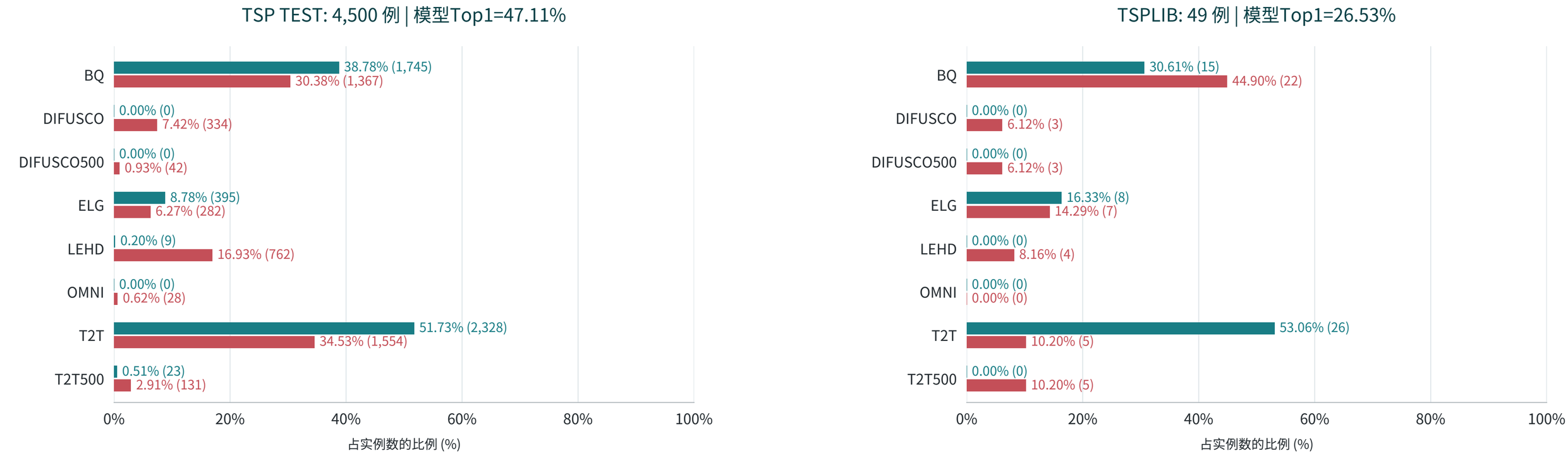
Normalization: new81000 TRAIN only; no TEST/LIB refit.

R89/B_muon_hybrid | TSP | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 41fa4ad8b590fa44

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/TSPLIB | native/FP64 ind



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

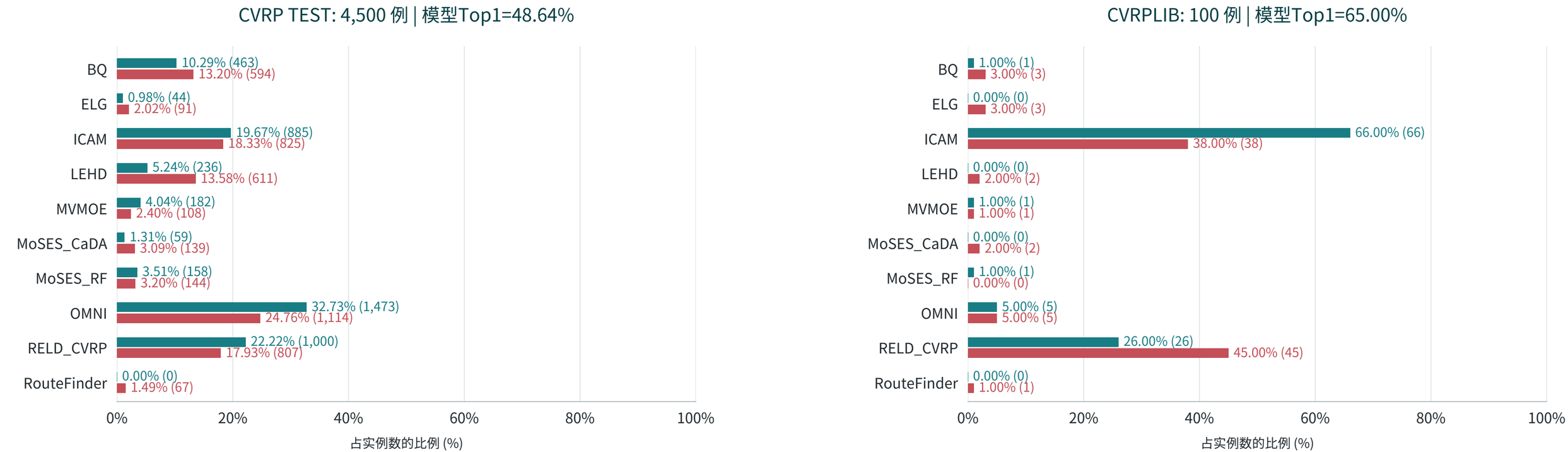
R89/B_muon_hybrid: stable global solver-ID score ties. Same R89 lock; LIB inference only; no training/reselection/Rxx mixing.

R89/B_muon_hybrid | CVRP | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 41fa4ad8b590fa44

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/CVRPLIB | native/FP64 ind



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

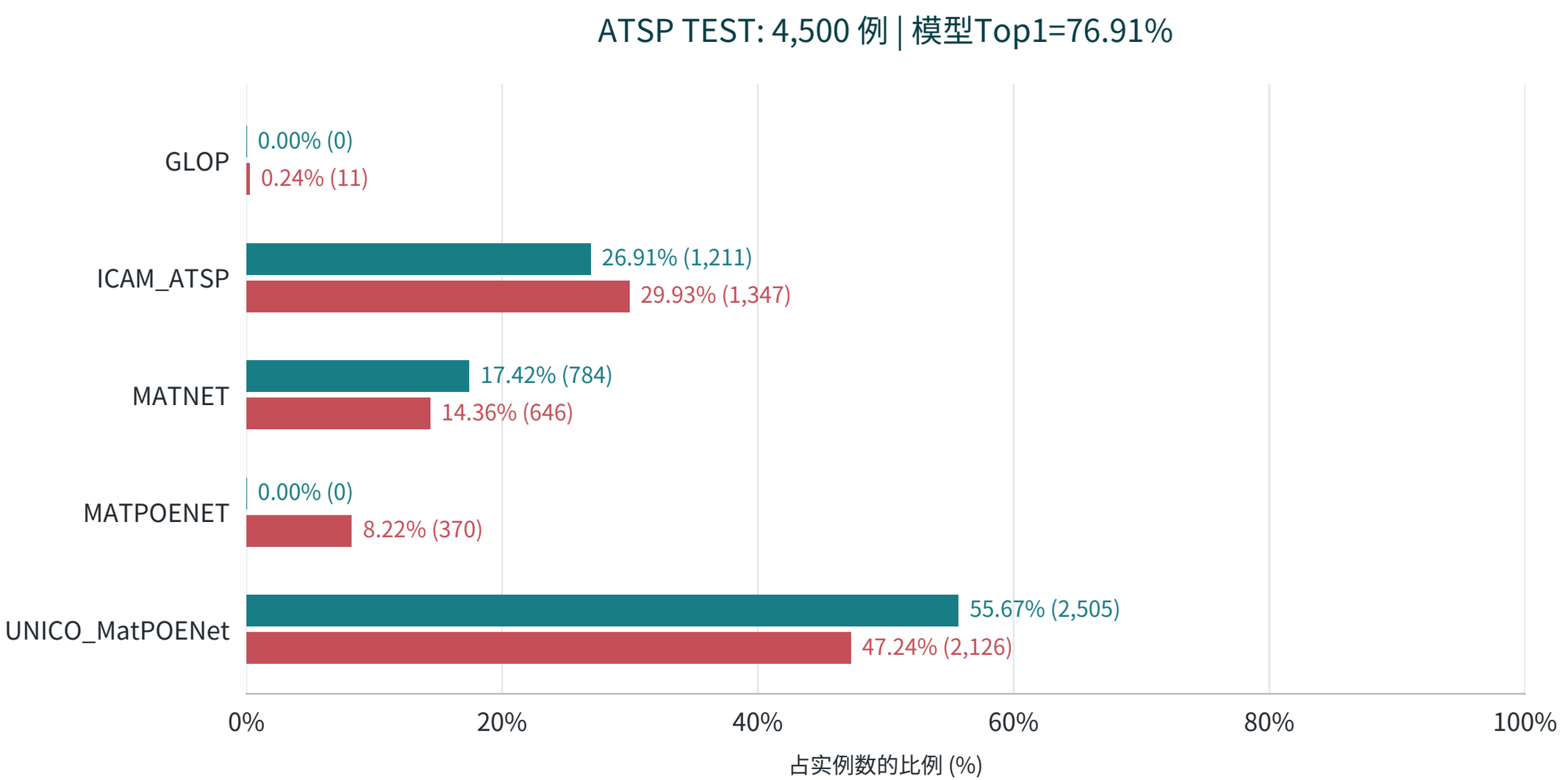
R89/B_muon_hybrid: stable global solver-ID score ties. Same R89 lock; LIB inference only; no training/reselection/Rxx mixing.

R89/B_muon_hybrid | ATSP | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 41fa4ad8b590fa44

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

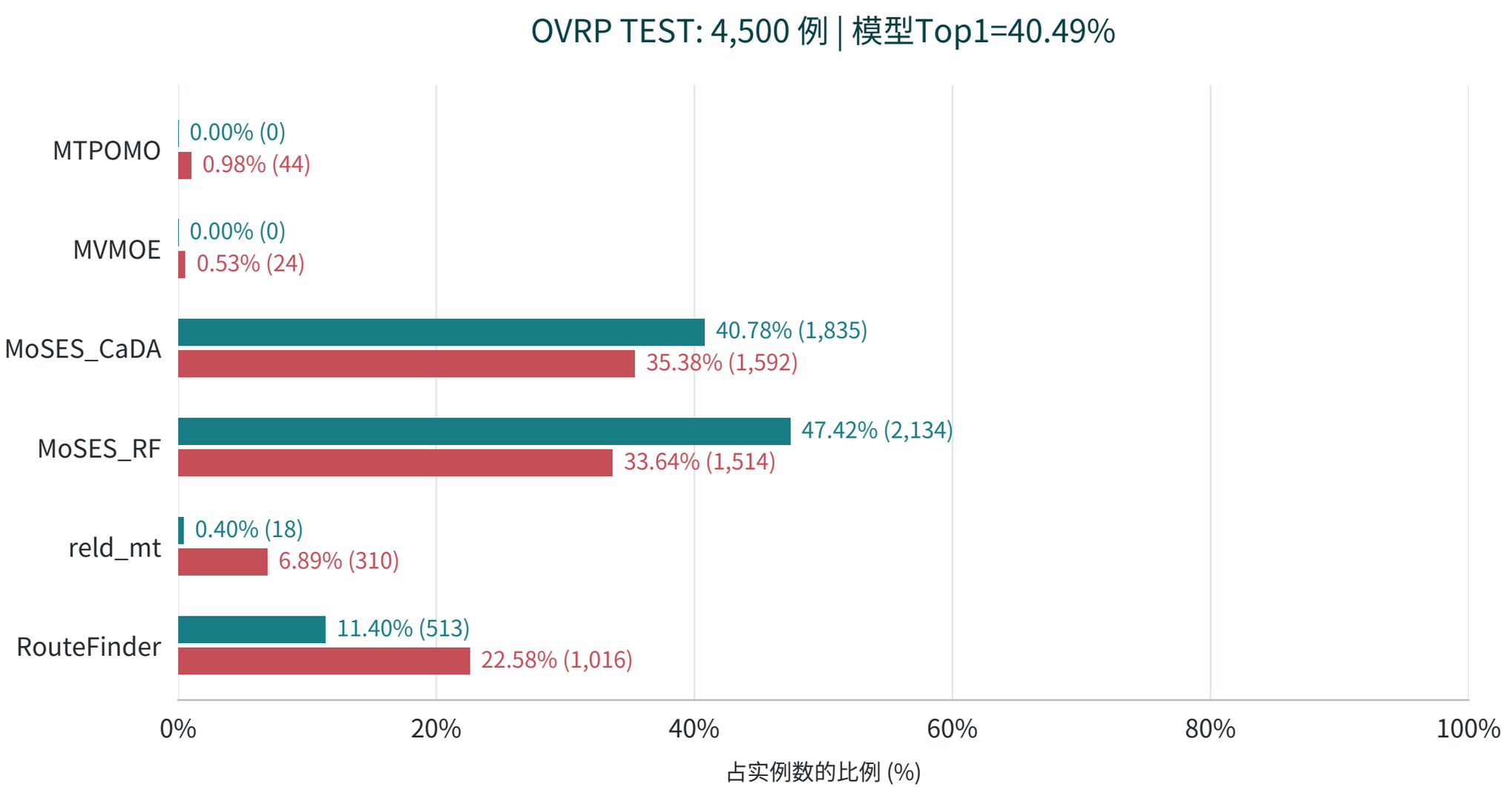
R89/B_muon_hybrid: stable global solver-ID score ties. Same R89 lock; LIB inference only; no training/reselection/Rxx mixing.

R89/B_muon_hybrid | OVRP | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 41fa4ad8b590fa44

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

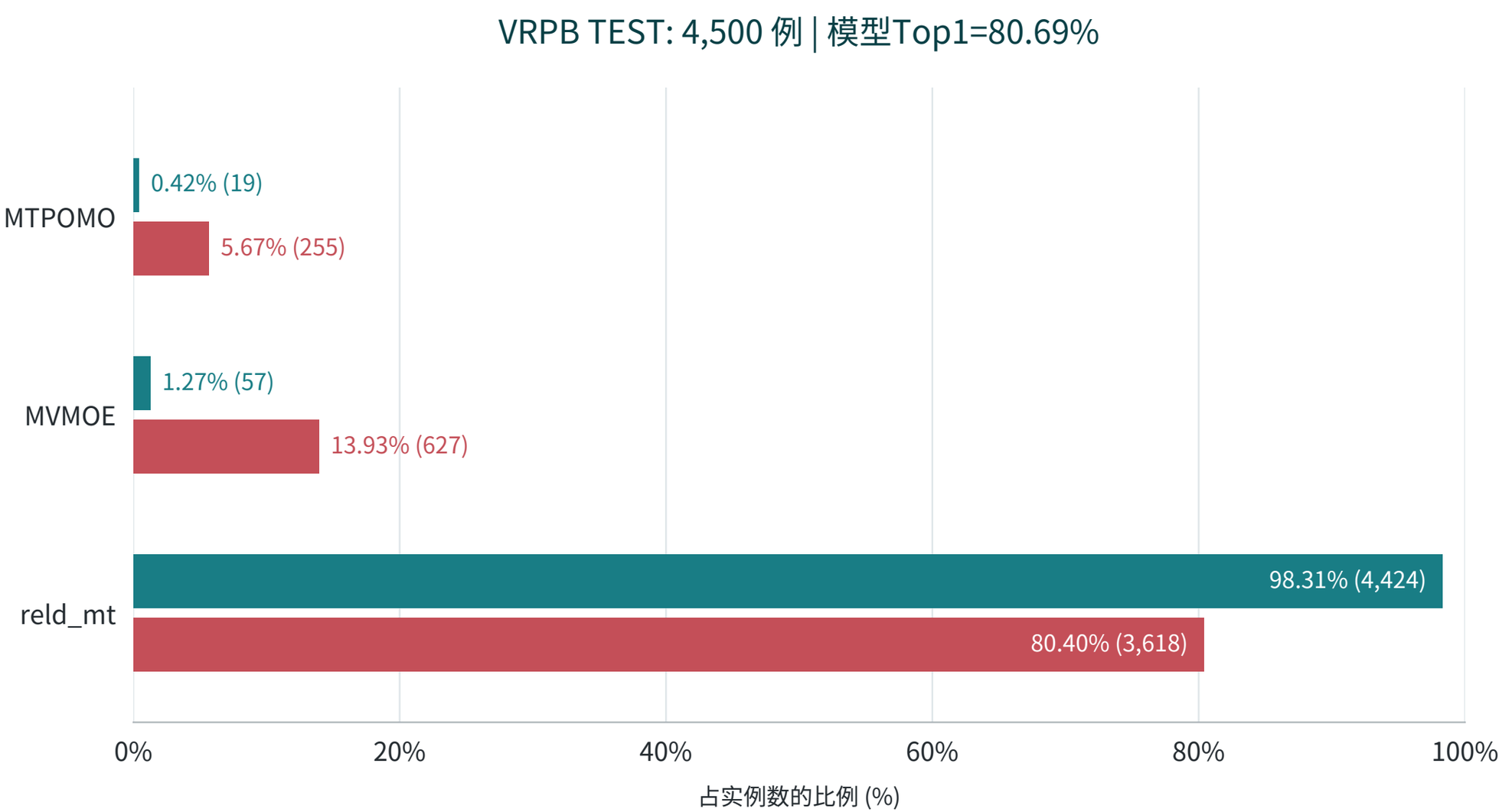
R89/B_muon_hybrid: stable global solver-ID score ties. Same R89 lock; LIB inference only; no training/reselection/Rxx mixing.

R89/B_muon_hybrid | VRPB | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 41fa4ad8b590fa44

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

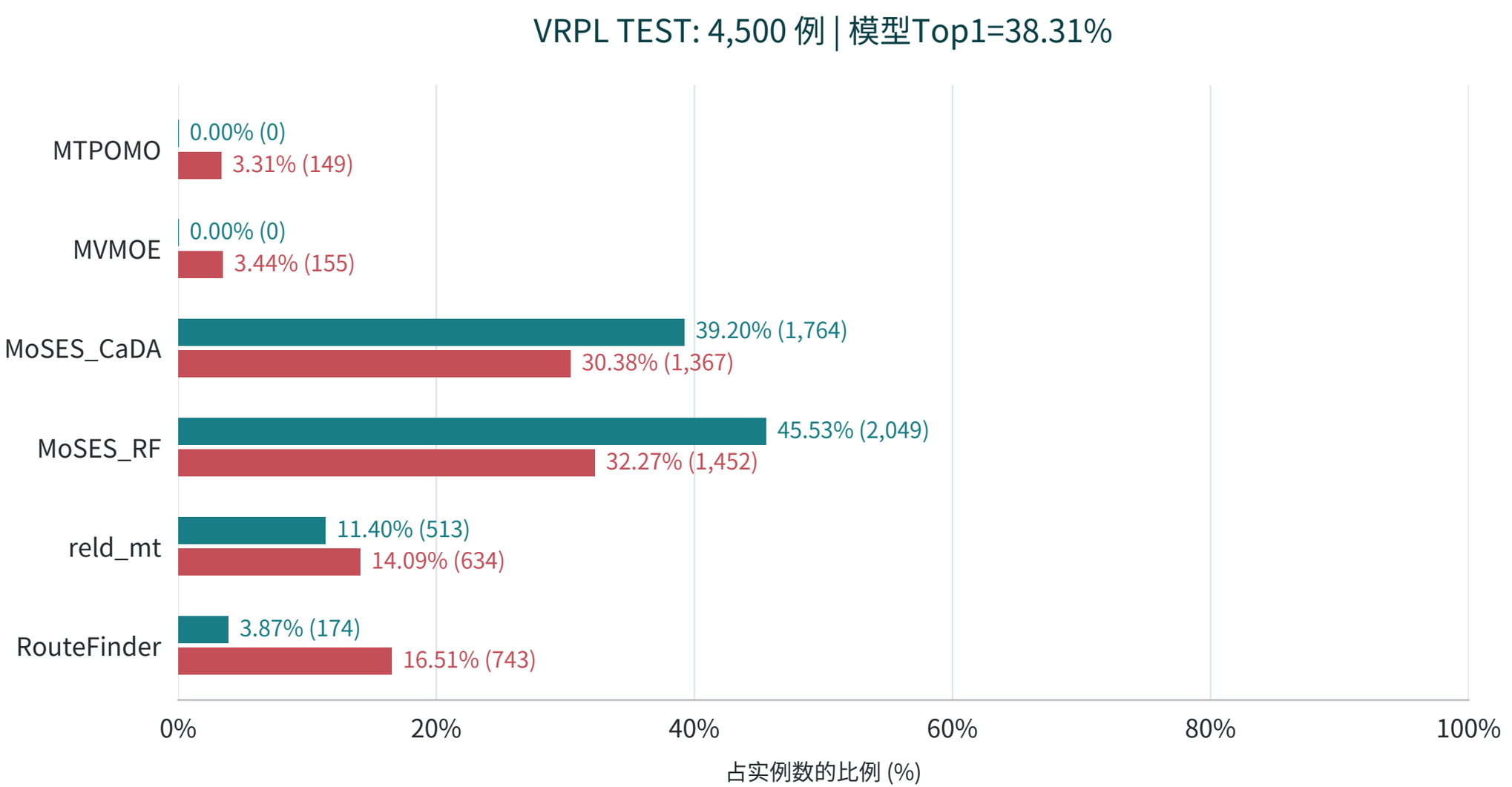
R89/B_muon_hybrid: stable global solver-ID score ties. Same R89 lock; LIB inference only; no training/reselection/Rxx mixing.

R89/B_muon_hybrid | VRPL | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 41fa4ad8b590fa44

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

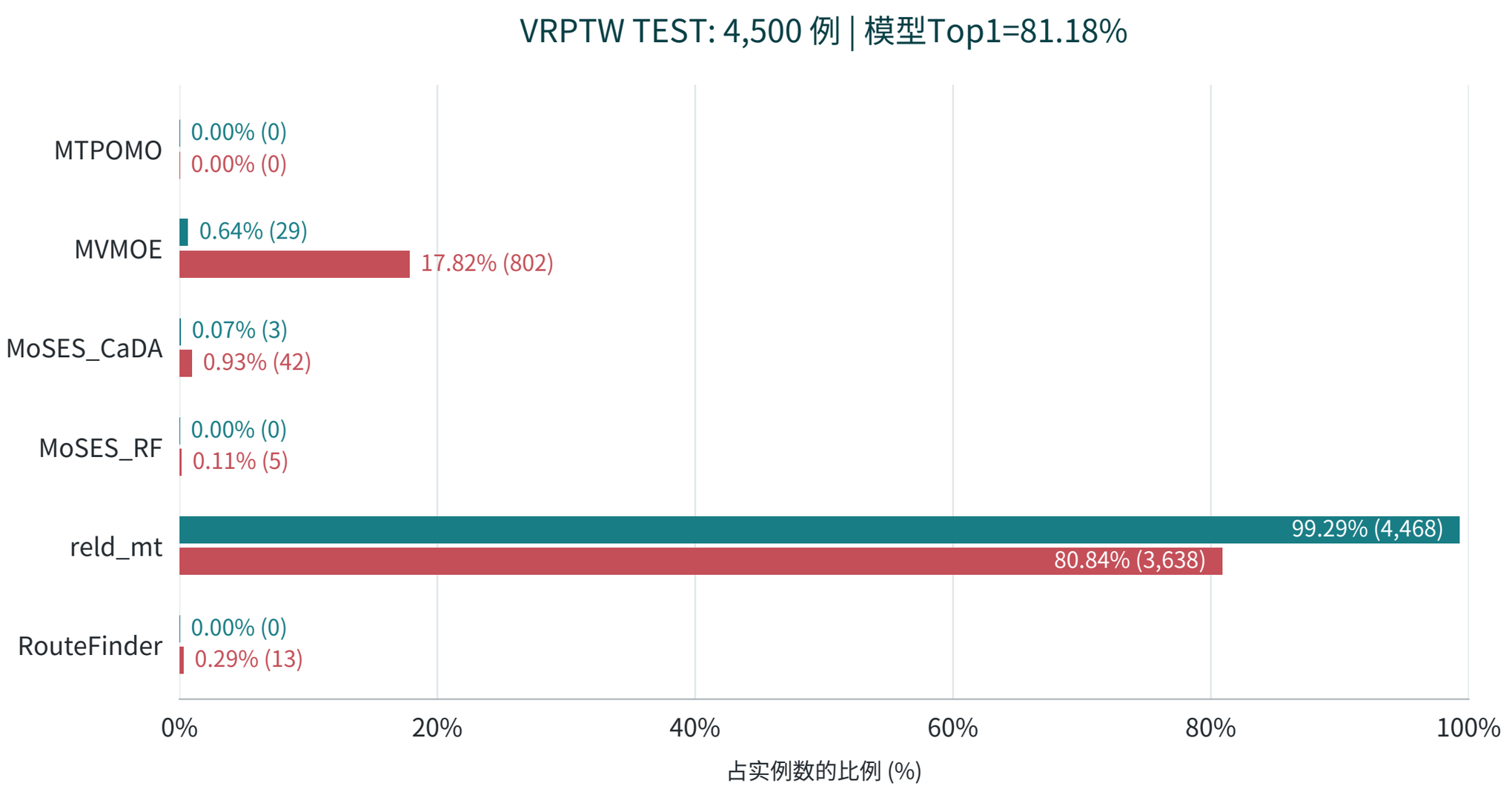
R89/B_muon_hybrid: stable global solver-ID score ties. Same R89 lock; LIB inference only; no training/reselection/Rxx mixing.

R89/B_muon_hybrid | VRPTW | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 41fa4ad8b590fa44

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

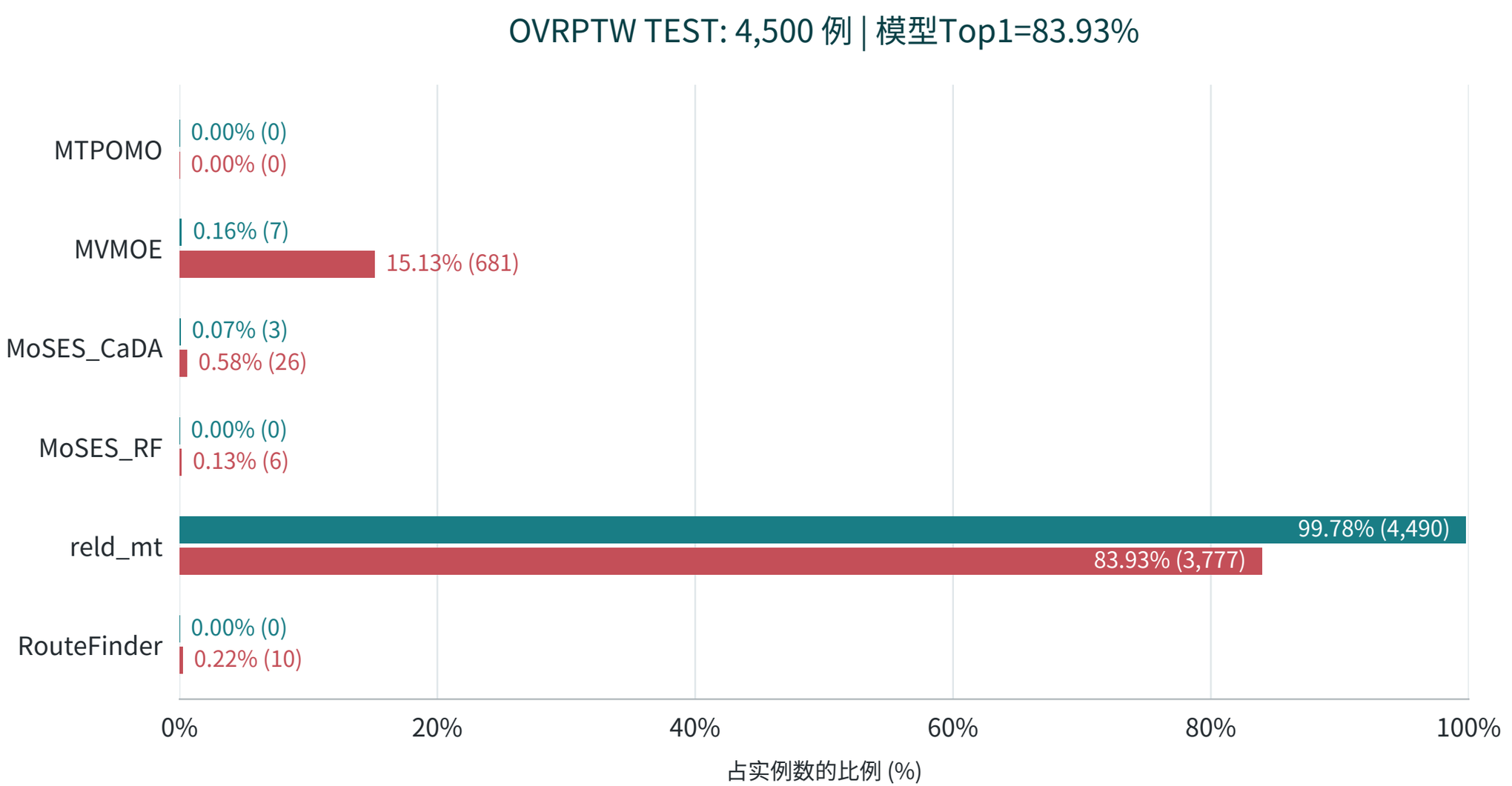
R89/B_muon_hybrid: stable global solver-ID score ties. Same R89 lock; LIB inference only; no training/reselection/Rxx mixing.

R89/B_muon_hybrid | OVRPTW | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 41fa4ad8b590fa44

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

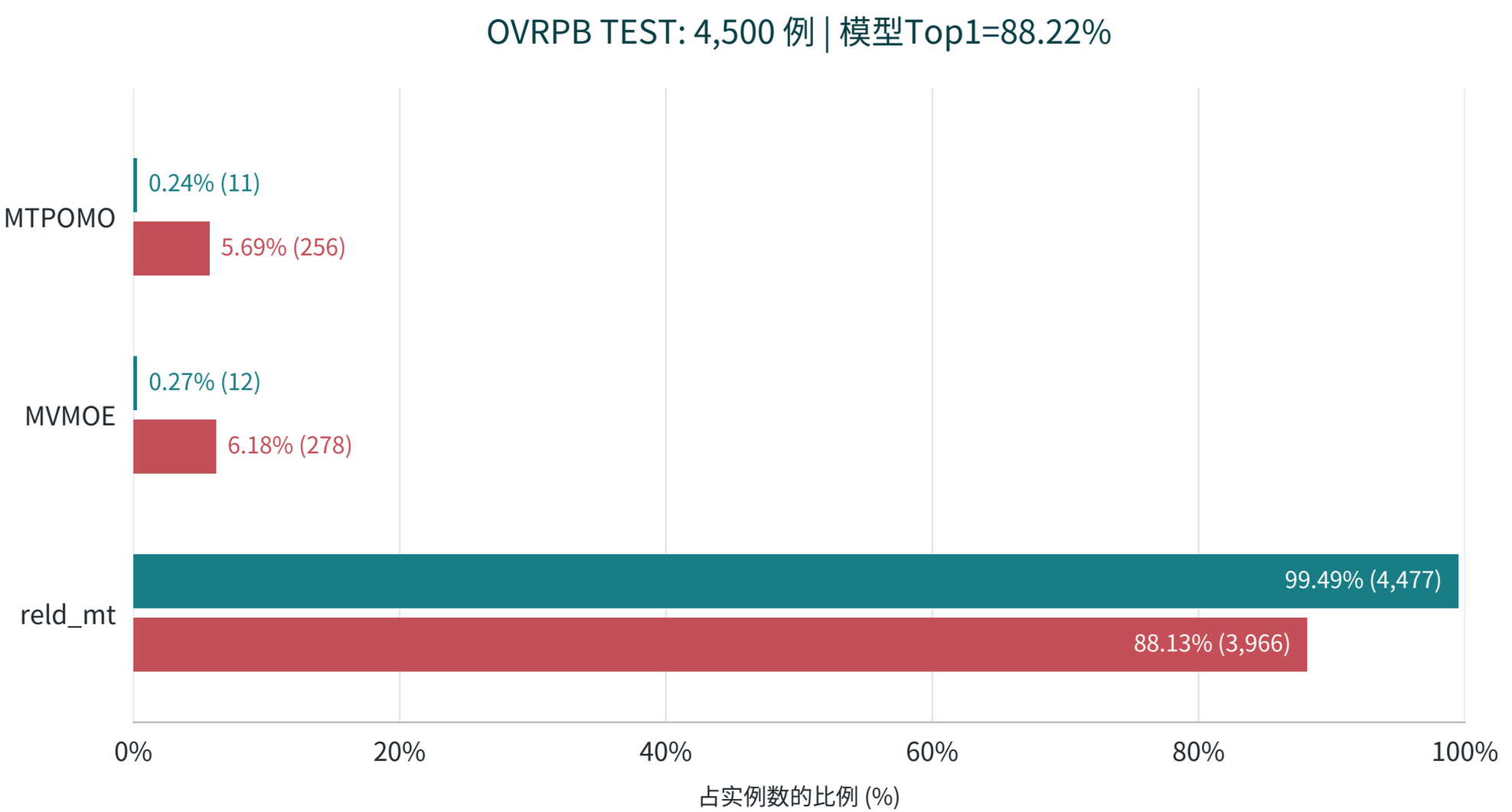
R89/B_muon_hybrid: stable global solver-ID score ties. Same R89 lock; LIB inference only; no training/reselection/Rxx mixing.

R89/B_muon_hybrid | OVRPB | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 41fa4ad8b590fa44

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

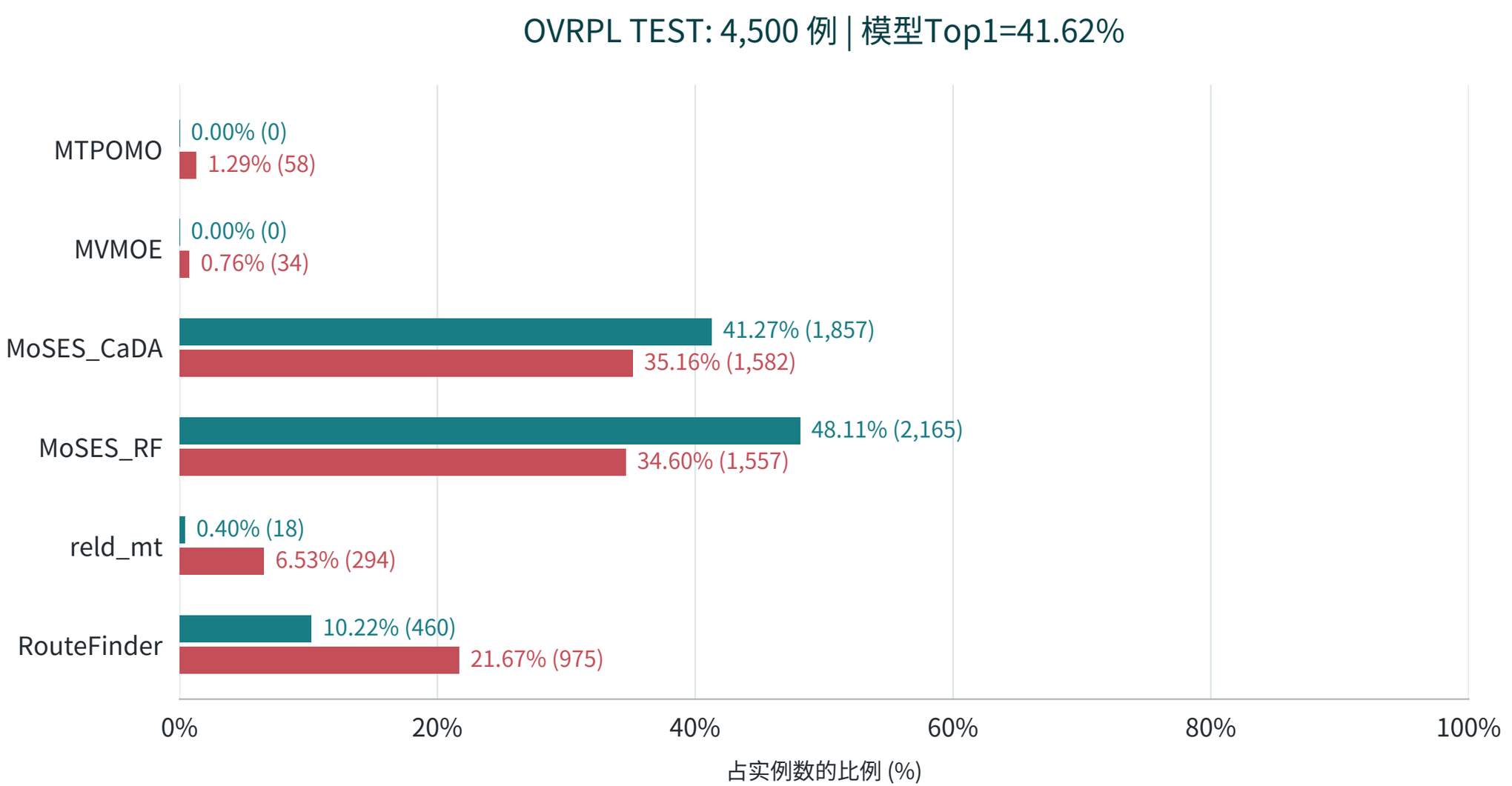
R89/B_muon_hybrid: stable global solver-ID score ties. Same R89 lock; LIB inference only; no training/reselection/Rxx mixing.

R89/B_muon_hybrid | OVRPL | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 41fa4ad8b590fa44

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

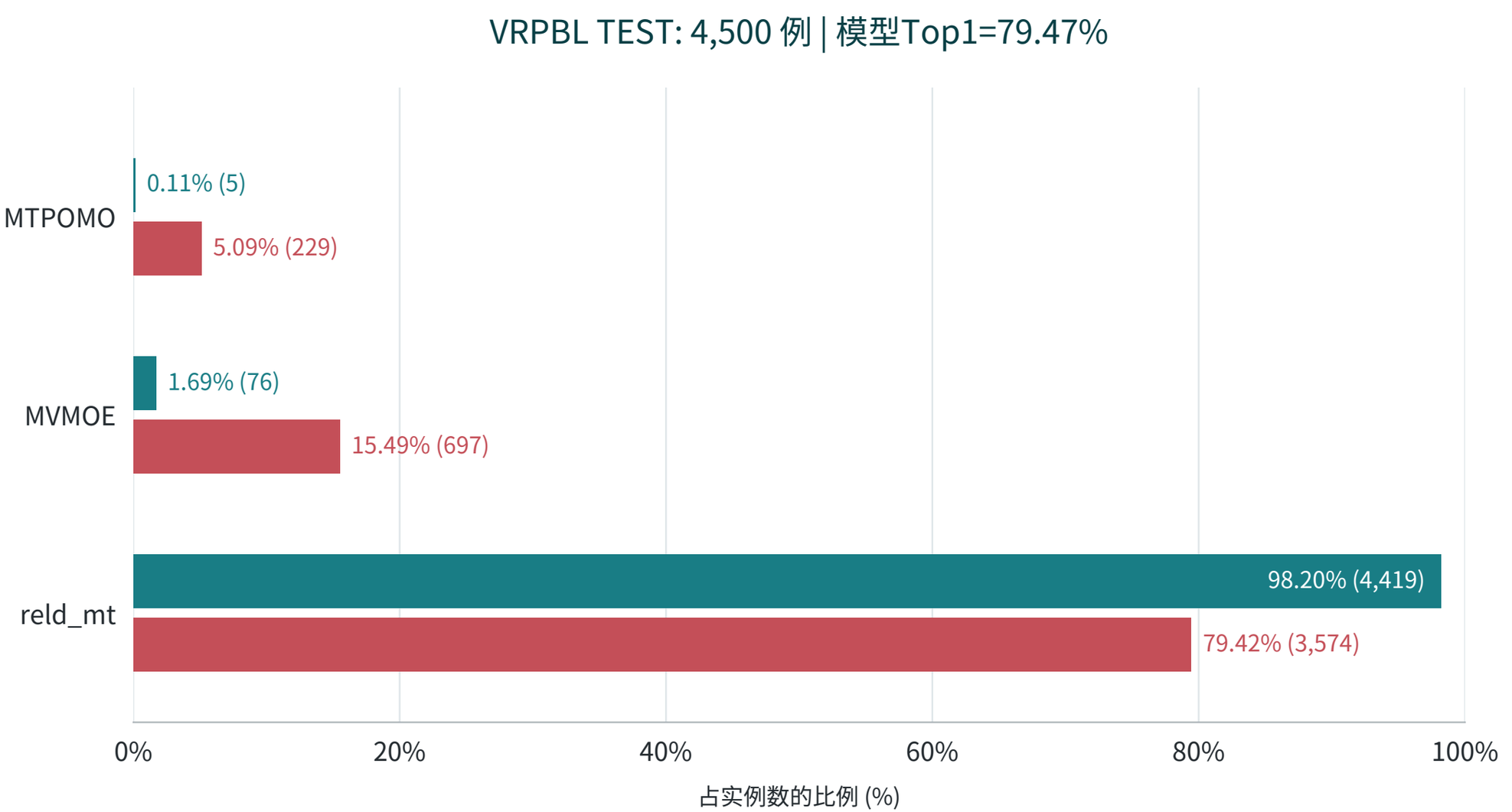
R89/B_muon_hybrid: stable global solver-ID score ties. Same R89 lock; LIB inference only; no training/reselection/Rxx mixing.

R89/B_muon_hybrid | VRPBL | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 41fa4ad8b590fa44

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

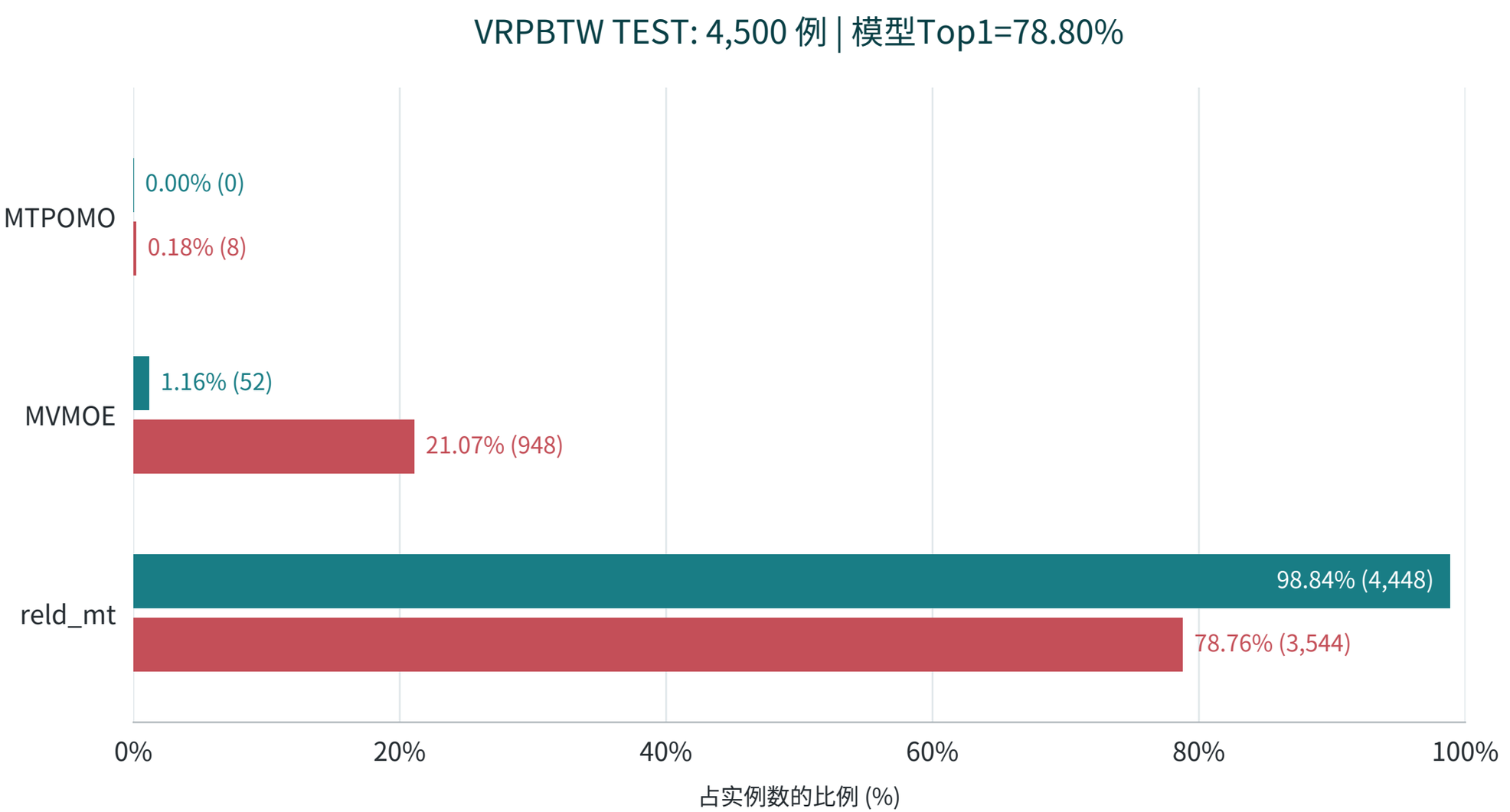
R89/B_muon_hybrid: stable global solver-ID score ties. Same R89 lock; LIB inference only; no training/reselection/Rxx mixing.

R89/B_muon_hybrid | VRPBTW | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 41fa4ad8b590fa44

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

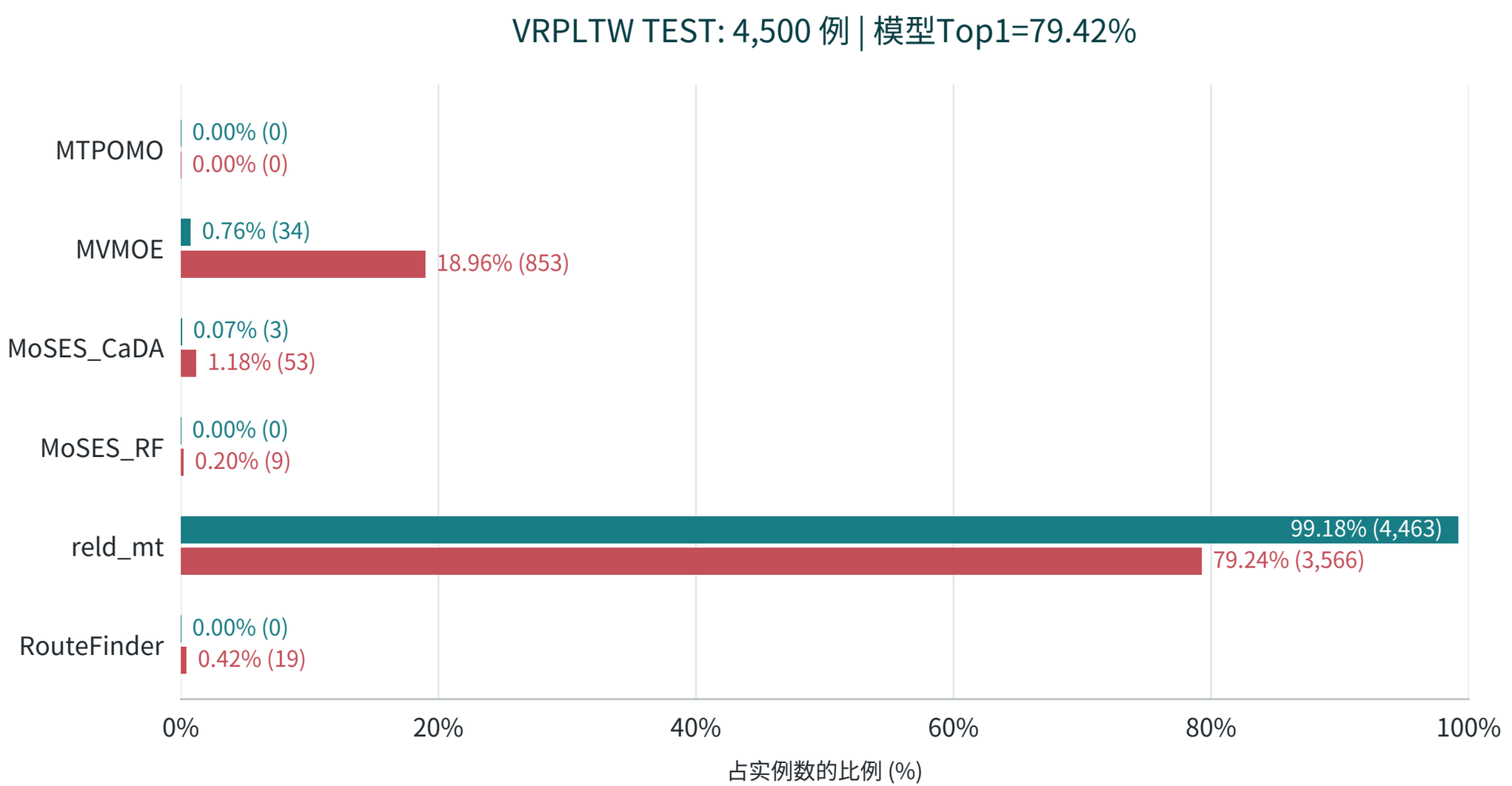
R89/B_muon_hybrid: stable global solver-ID score ties. Same R89 lock; LIB inference only; no training/reselection/Rxx mixing.

R89/B_muon_hybrid | VRPLTW | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 41fa4ad8b590fa44

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

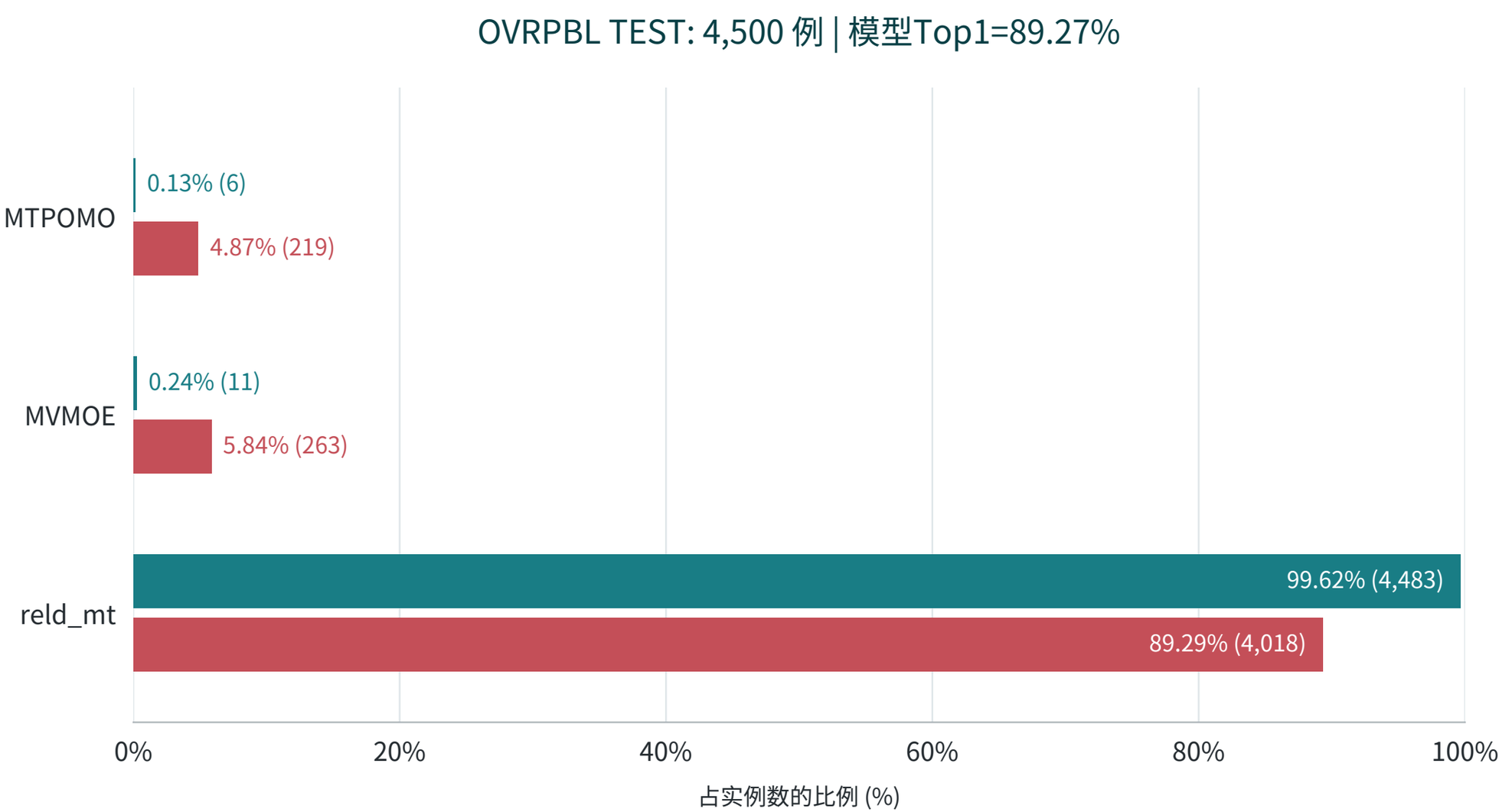
R89/B_muon_hybrid: stable global solver-ID score ties. Same R89 lock; LIB inference only; no training/reselection/Rxx mixing.

R89/B_muon_hybrid | OVRPBL | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 41fa4ad8b590fa44

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

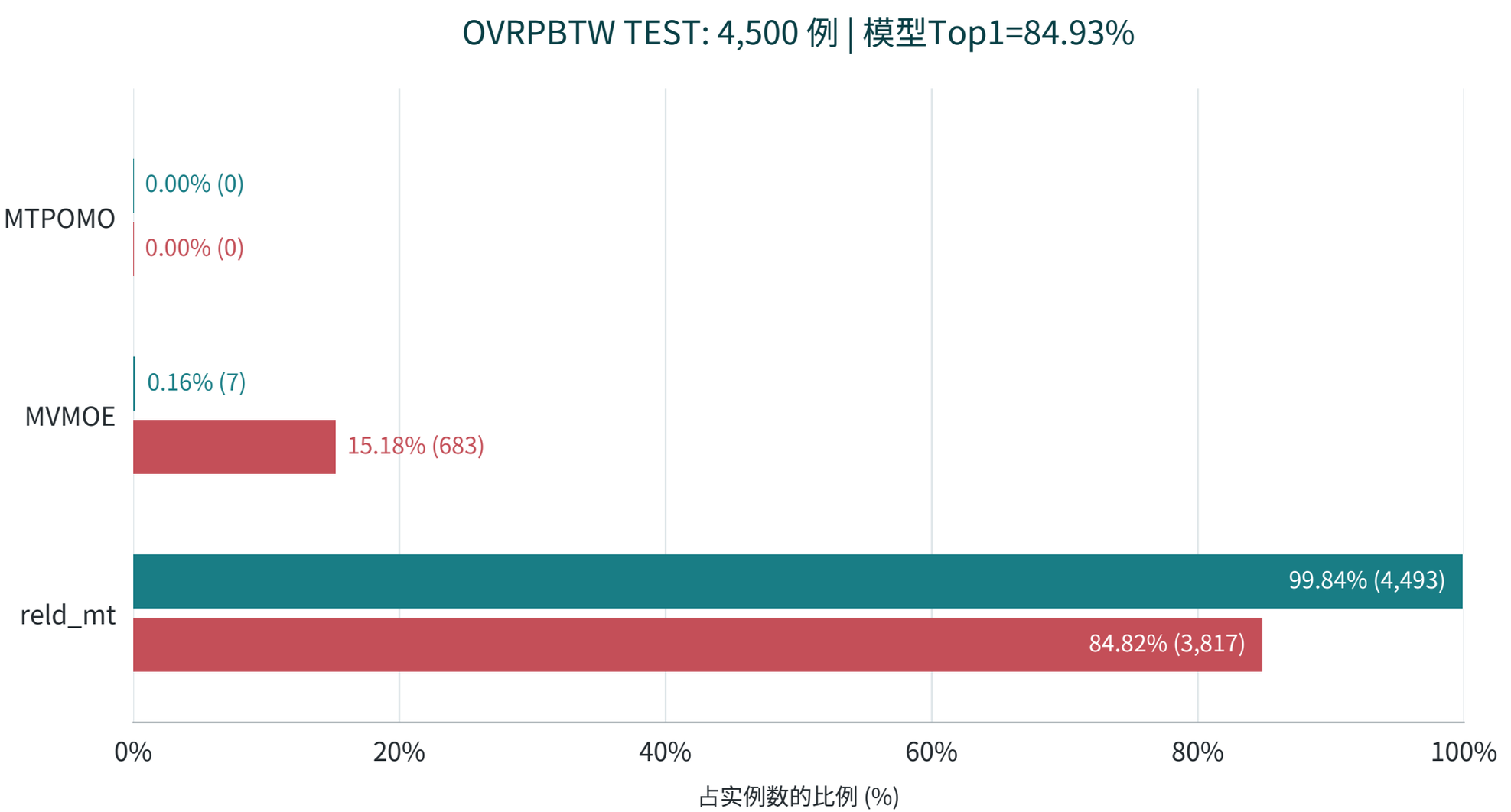
R89/B_muon_hybrid: stable global solver-ID score ties. Same R89 lock; LIB inference only; no training/reselection/Rxx mixing.

R89/B_muon_hybrid | OVRPBTW | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 41fa4ad8b590fa44

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

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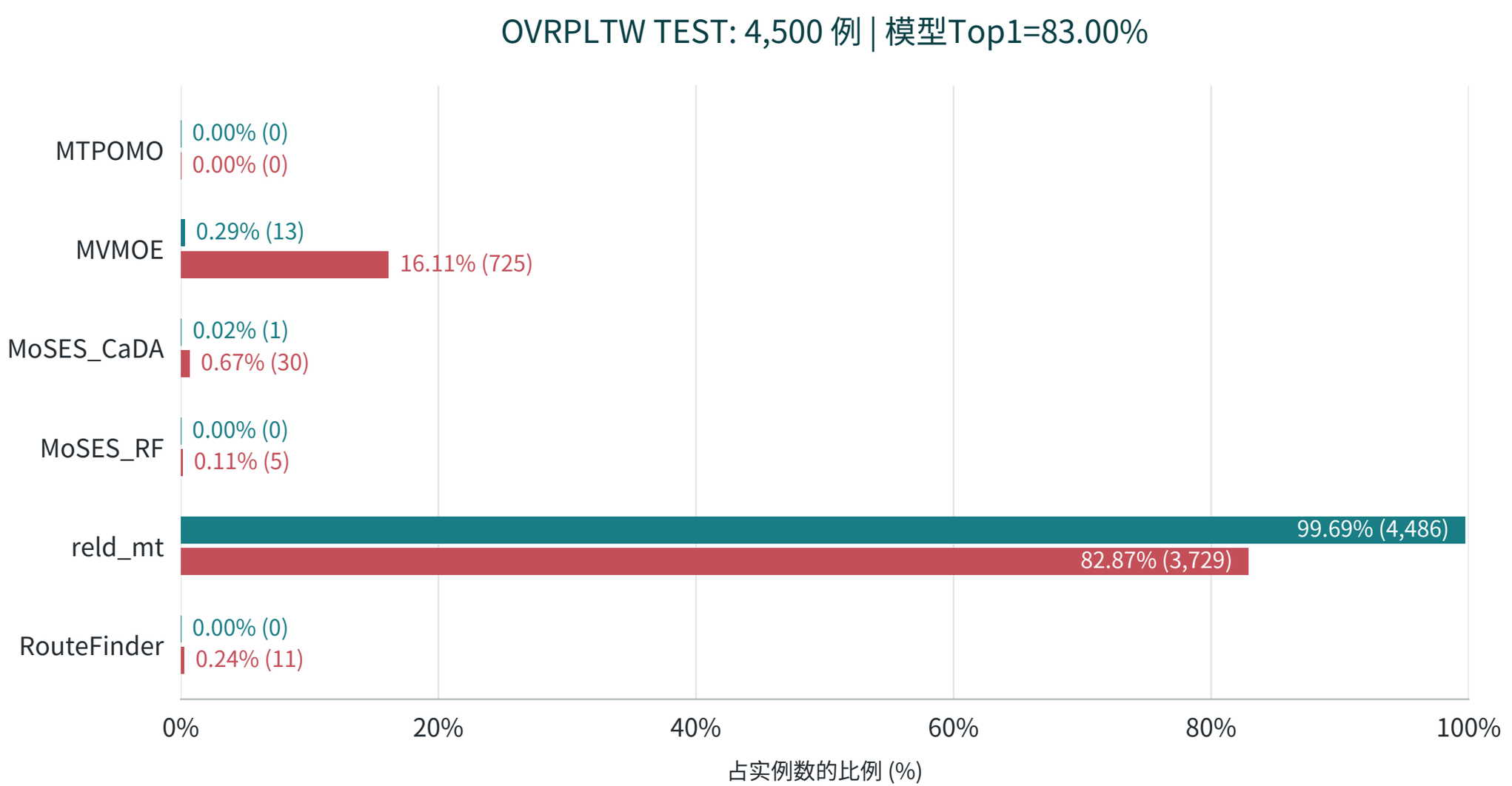
R89/B_muon_hybrid: stable global solver-ID score ties. Same R89 lock; LIB inference only; no training/reselection/Rxx mixing.

R89/B_muon_hybrid | OVRPLTW | 方法分布

同一锁定模型: epoch 25 / update 36000 / SHA256 41fa4ad8b590fa44

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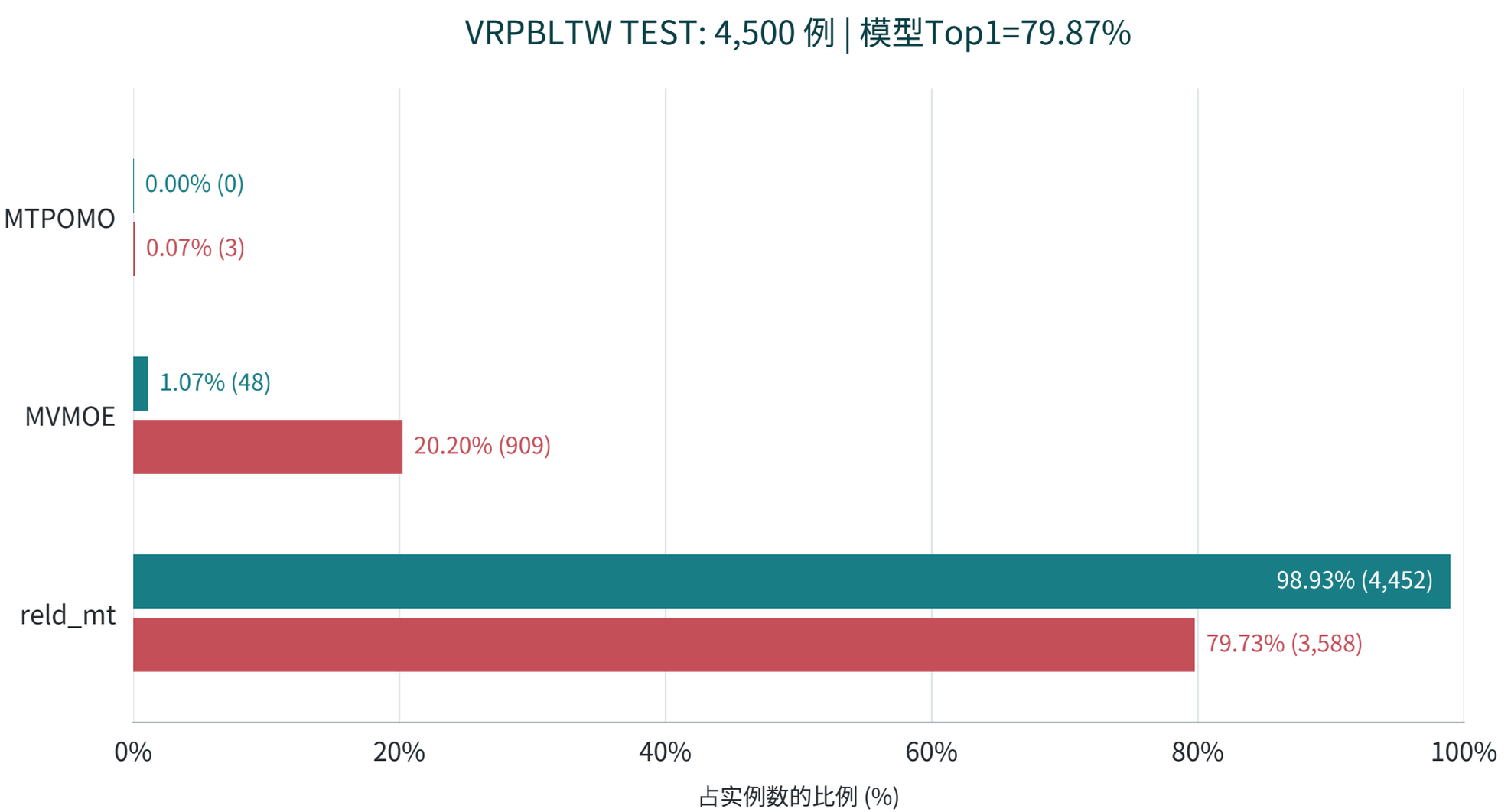
R89/B_muon_hybrid: stable global solver-ID score ties. Same R89 lock; LIB inference only; no training/reselection/Rxx mixing.

R89/B_muon_hybrid | VRPBLTW | 方法分布

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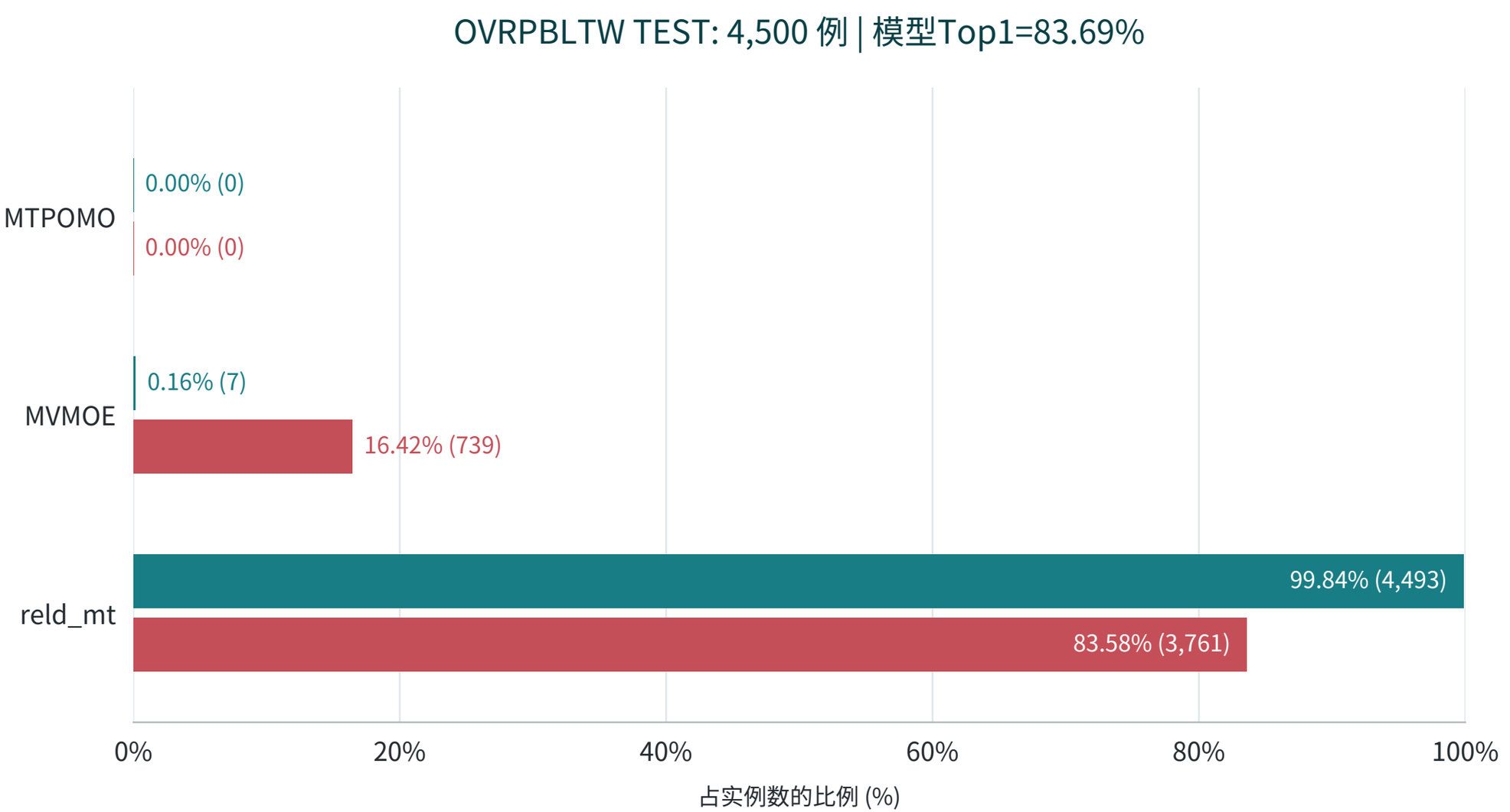
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